

Project Initialization and Planning Phase

Date	6 July 2026
Team ID	xxxxxx
Project Name	Credit Card Fraud Detection System using Machine Learning
Maximum Marks	2 Marks

Brainstorming:

Brainstorming helps a team generate and organize ideas before development begins. For our Credit Card Fraud Detection System, our team brainstormed the project scope, objective, and expected benefits of using machine learning to identify fraudulent transactions. This session helped align the team on a shared direction before moving into detailed planning and development.

Project	Objective	Benefits
Credit Card Fraud Detection System using Machine Learning.	Develop a machine learning web application to classify credit card transactions as fraudulent or legitimate.	Real-time prediction, reduced manual review effort, and improved fraud detection accuracy.

Key Brainstorming Ideas

- Use a supervised machine learning model to classify each transaction as fraudulent or legitimate in real time.
- Train and compare multiple algorithms, such as Logistic Regression, Decision Tree, Random Forest, and XGBoost, to select the best-performing model.
- Build a simple web interface where users can enter transaction details and instantly view the prediction result.
- Reduce dependency on manual transaction review, allowing bank staff to focus on confirmed high-risk cases.
- Provide a confidence score alongside the prediction to help users and analysts interpret the result.
- Deploy the solution on the cloud so it is accessible through a public link without additional setup.

Conclusion

The brainstorming session helped our team define a clear project scope, objective, and expected benefits for the Credit Card Fraud Detection System. These ideas laid the foundation for identifying user problem statements and planning the machine learning solution in the subsequent stages of the project.